

1. suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

a) what is the mean and median

$$\text{mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{1}{27} (13 + 15 + 16 + 16 + 19 + 20 + 20 + 21 + 22 + 22 + 25 + 25 + 25 + 25 + 30 + 33 + 33 + 35 + 35 + 35 + 35 + 36 + 40 + 45 + 46 + 52 + 70)$$

$$= \frac{1}{27} (809) = 29.962 = 30 \text{ (approx)}$$

$$\text{Median} = \frac{n+1}{2} = \frac{28}{2} = 14$$

$$= 25 \text{ (14th value is the Median value)}$$

b) what is the mode

$\therefore$ 25 and 35 are repeating for 4 times

Therefore, It is bimodal.

$$\text{c) Midrange} = \frac{\text{maximum} + \text{minimum}}{2}$$

$$= \frac{70 + 13}{2} = \frac{83}{2} = 41.5$$

d) first Quartile (Q_1) and Third Quartile (Q_3)

$Q_1 = 13, 15, 16, 16, 19, 20, \underline{20}, 21, 22, 22, 25, 25, 25$

$$\text{median} = \frac{n+1}{2} = \frac{13+1}{2} = \frac{14}{2} = 7$$

$$Q_1 = 20$$

$Q_3 = 30, 33, 33, 35, 35, 35, \underline{35}, 36, 40, 45, 46, 52, 7$

$$Q_3 = 35$$

e) five-number summary

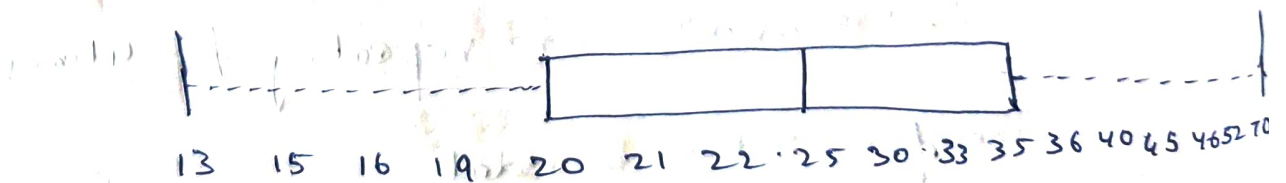
$$IQR = Q_3 - Q_1 = 35 - 20 = 15$$

e) five-number summary

$$= \min, Q_1, \text{Median}, Q_3, \max$$

$$= 13, 20, 25, 35, 70$$

f) Boxplot



2. Given the following measurements for the.

Variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

standardize the variable by the following:

compute the mean absolute deviation of age.

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{1}{10} (18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28)$$

$$= \frac{1}{10} (330) = 33$$

Age(x)	$ x - \bar{x} $
18	$18 - 33 = 15$
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5
	88

$$= \frac{\sum |x - \bar{x}|}{n}$$

$$= \frac{88}{10} = 8.8$$

3) suppose that the data for analysis includes the attribute age. The age values for the data tuples are, (in increasing order)

13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

use smoothing by bin means to smooth the above data, using a bin depth of 3.

Illustrate your steps

$$\begin{aligned} \text{Total no. of bin} &= \frac{\text{Total}}{\text{bin depth}} \\ &= \frac{27}{3} = 9 \end{aligned}$$

Bin	Original values	Bin Mean	Smoothed values
B1	13, 15, 16	14.67	14.67, 14.67, 14.67
B2	16, 19, 20	18.3	18.3, 18.3, 18.3
B3	20, 21, 22	21	21.00, 21.00, 21.00
B4	22, 25, 25	24	24.00, 24.00, 24.00
B5	25, 25, 30	26.67	26.67, 26.67, 26.67
B6	33, 33, 35	33.67	33.67, 33.67, 33.67
B7	35, 35, 35	35	35.00, 35.00, 35.00
B8	36, 40, 45	40.3	40.33, 40.33, 40.33
B9	46, 52, 70	56	56.00, 56.00, 56.00

a) Min-Max Normalization

$$= \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$= \frac{x - 200}{1000 - 200} = \frac{x - 200}{800}$$

Original value (x)	Calculation	(x')
200	$\frac{200 - 200}{800}$	0.000
300	$\frac{300 - 200}{800}$	0.125
400	$\frac{400 - 200}{800}$	0.250
600	$\frac{600 - 200}{800}$	0.500
1000	$\frac{1000 - 200}{800}$	1.000

4. Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

a) min-max normalization by setting min=0 and max=1 b) Z-score normalization

$$b) \quad z = \frac{x - \mu}{\sigma}$$

$$\text{Mean} = \frac{1}{5} (200 + 300 + 400 + 600 + 1000)$$

$$(c) \quad \frac{2500}{5} = 500$$

Standard deviation

x	$x - \mu$	$(x - \mu)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000

$$\sum (x - \mu)^2 = 400000$$

Variance :

$$\sigma^2 = \frac{400000}{5} = 80000$$

Standard deviation :

$$\sigma = \sqrt{80000} \approx 282.84$$

compute z-scores

original value	calculation	z-score
200	$(200 - 500) / 282.84$	-1.06
300	$(300 - 500) / 282.84$	-0.71
400	$(400 - 500) / 282.84$	-0.35
600	$(600 - 500) / 282.84$	0.35
1000	$(1000 - 500) / 282.84$	1.77

$-1.06, -0.71, -0.35, 0.35, 1.77$

5. Suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods:
- (a) equal-frequency partitioning
 - (b) equal width partitioning
 - (c) clustering

Total no. of records = 12

Total no. of bins = 3

a) Records: Equal frequency partitioning

$$\text{records' bins} = \frac{12}{3} = 4$$

Bin

Values

Bin 1

5, 10, 11, 13

Bin 2

15, 35, 50, 55

Bin 3

72, 92, 204, 215

b) equal-width partitioning.

$$\text{width} = \frac{\text{maximum} - \text{minimum}}{\text{no. of bin}}$$

$$= \frac{215 - 5}{3} = \frac{210}{3} = 70$$

The intervals are

Bin 1: 5-74

Bin 2: 75-144

Bin 3: 145-215

Now plot

Bin	Interval	values
Bin 1	5-74	5, 10, 11, 13, 15, 35, 50, 55
Bin 2	75-144	72, 92
Bin 3	145-215	204, 215

c) Clustering

cluster(Bin)	values
cluster 1	5, 10, 11, 13, 15
cluster 2	35, 50, 55, 72, 92
cluster 3	204, 215

6. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result

age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	26.1	31.9	46.2

$$\text{Mean of age} = \frac{836}{18} = 46.44$$

$$\text{Mean of \% fat} = \frac{518.1}{18} = 28.78$$

$$\text{Median of age} = \frac{50+52}{2} = \frac{102}{2} = 51$$

$$\text{Median of \% fat} = 30.70$$

Standard Deviation of Age

no. of observation, $n = 18$

$$\text{Mean age } \bar{x} = 46.44$$

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n-1}$$

$$\sum (x - \bar{x})^2 = \underline{3369.94}$$

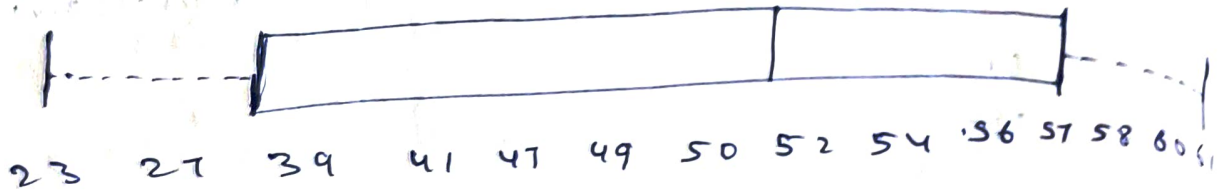
$$\text{sample variance} = \frac{3369.94}{17} = 198.23$$

$$\text{Standard deviation} = \sqrt{198.23} = 14.08$$

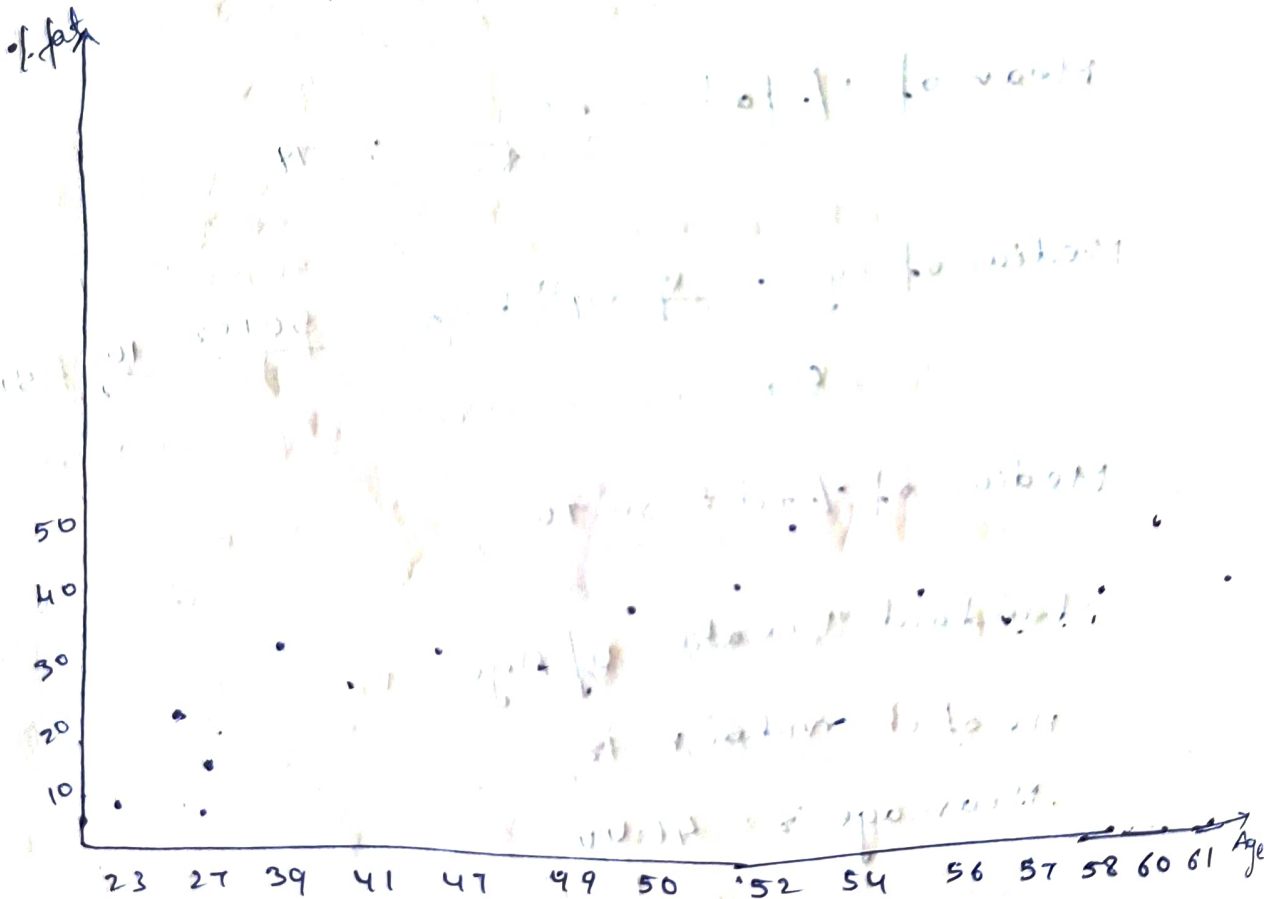
b) five number Summary

min, Q_1 , median, Q_3 , max

23, 36, 51, 57, 61



c) Scatter Plot



d) Z Normalization

$$Z = \frac{X - \bar{x}}{S}$$

Age z

$$23 \quad \frac{23-46.44}{12.85} = -1.82$$

$$23 \quad -1.82$$

$$27 \quad -1.51$$

$$27 \quad -1.51$$

$$39 \quad -0.57$$

$$41 \quad -0.42$$

$$47 \quad -0.643$$

$$49 \quad 0.199$$

$$50 \quad 0.27$$

$$52 \quad (1.0043) - (1.0029) = 0.0014$$

$$54 \quad 0.58$$

$$54 \quad 0.58$$

$$56 \quad 0.74$$

$$57 \quad 0.82$$

$$58 \quad 0.90$$

$$58 \quad 0.90$$

$$60 \quad 1.05$$

$$61 \quad 1.13$$

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$\sum x = 836$$

$$\sum y = 518.1$$

$$\sum x^2 = 698,296$$

$$\sum x^2 = 41,798$$

$$\sum y^2 = 16,368.59$$

$$\sum xy = 25,763.2$$

$$r = \frac{18(25,763.2) - (836)(518.1)}{\sqrt{[18(41,798) - (698,296)][18(16,368.59) - 268,427.6]}}$$

$$= \frac{463,737.6 - 433,131.6}{(53,468)(26,196.4)}$$

$$\frac{30,606}{30,606}$$

$$= 0.82$$